Describe ratio relationships

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: There are 8 red and 12 blue counters. Write red to blue as a ratio. Copy or complete the first step.

2. Explain the ratio 5:2 in a sentence.

3. Complete the equivalent ratio 7:4 = 28:__.

4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: There are 8 red and 12 blue counters. Write red to blue as a ratio. Copy or complete the first step.
   Answer 8:12, which simplifies to 2:3.

2. Explain the ratio 5:2 in a sentence.
   Answer For every 5 of the first quantity, there are 2 of the second.

3. Complete the equivalent ratio 7:4 = 28:__.
   Answer 16.

4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.
   Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.

Describe ratio relationships

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. There are 8 red and 12 blue counters. Write red to blue as a ratio.

2. Explain the ratio 5:2 in a sentence.

3. Complete the equivalent ratio 7:4 = 28:__.

4. Solve this independently and show every stage: There are 8 red and 12 blue counters. Write red to blue as a ratio.

5. Use a second method or representation for: Explain the ratio 5:2 in a sentence.

6. Check this result using an inverse, estimate or known fact: Complete the equivalent ratio 7:4 = 28:__.

7. Prove or explain your reasoning: Simplify 36:48.

8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Complete the equivalent ratio 7:4 = 28:__.

Teacher answers and checking prompts

1. There are 8 red and 12 blue counters. Write red to blue as a ratio.
   Answer 8:12, which simplifies to 2:3.

2. Explain the ratio 5:2 in a sentence.
   Answer For every 5 of the first quantity, there are 2 of the second.

3. Complete the equivalent ratio 7:4 = 28:__.
   Answer 16.

4. Solve this independently and show every stage: There are 8 red and 12 blue counters. Write red to blue as a ratio.
   Answer 8:12, which simplifies to 2:3.

5. Use a second method or representation for: Explain the ratio 5:2 in a sentence.
   Answer Expected result: For every 5 of the first quantity, there are 2 of the second. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Complete the equivalent ratio 7:4 = 28:__.
   Answer Expected result: 16. A valid check must be shown.

7. Prove or explain your reasoning: Simplify 36:48.
   Answer 3:4. The explanation must show why the method works.

8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.
   Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Complete the equivalent ratio 7:4 = 28:__.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Describe ratio relationships

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ___________________

1. Prove or explain your reasoning: Simplify 36:48.

2. Prove the answer to this question, rather than only calculating it: There are 8 red and 12 blue counters. Write red to blue as a ratio.

3. Solve this in two different ways and compare them: Explain the ratio 5:2 in a sentence.

4. Work backwards from the answer to reconstruct the question: 16.

5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.

6. Create a new example that tests the same idea as: Complete the equivalent ratio 7:4 = 28:__.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Simplify 36:48.
   Answer 3:4. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: There are 8 red and 12 blue counters. Write red to blue as a ratio.
   Answer Expected result: 8:12, which simplifies to 2:3. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Explain the ratio 5:2 in a sentence.
   Answer Expected result: For every 5 of the first quantity, there are 2 of the second. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 16.
   Answer One valid reconstruction is: Complete the equivalent ratio 7:4 = 28:__. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.
   Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.

6. Create a new example that tests the same idea as: Complete the equivalent ratio 7:4 = 28:__.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.